

5.3 THE QUANTITY THEORY OF MONEY (QTM)

MARKING SCHEME

1990/AL/II/14 C	1995/AL/II/18 D	2000/AL/II/22 D	2005/AL/II/21 D (77%)	2014/DSE/I/37 D (56%)
1990/AL/II/26 C	1996/AL/II/14 A	2001/AL/II/23 C	2007/AL/II/11 B (77%)	2015/DSE/I/33 A (58%)
1991/AL/II/14 A	1996/AL/II/20 C	2002/AL/II/19 C (52%)	2008/AL/II/18 D (61%)	2016/DSE/I/34 D (34%)
1991/AL/II/16 A	1997/AL/II/12 B	2002/AL/II/23 B (73%)	2009/AL/II/22 C (71%)	2017/DSE/I/36 C (64%)
1992/AL/II/20 B	1997/AL/II/22 D	2003/AL/II/11 D	2011/AL/II/24 A (59%)	2018/DSE/I/37 B (72%)
1994/AL/II/17 D	1998/AL/II/14 C	2004/AL/II/19 C (44%)	2012/AL/II/16 C (69%)	2019/DSE/I/35 A
1994/AL/II/22 D	1999/AL/II/14 D	2005/AL/II/14 C (74%)	2012/AL/II/20 B (48%)	2021/DSE/I/38 A
1994/AL/II/24 B	2000/AL/II/16 D	2005/AL/II/18 A (56%)	2013/DSE/I/32 B (74%)	2023/DSE/I/37 B
2024/DSE/II/37 D	2025/DSE/II/39 B			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/AL/II/8(a)(ii)

It all depends.

If the growth rate of the money supply approximately equals the growth rate of real output, then the inflationary consequence will be minimal.

However, if the growth rate of the money supply exceeds the growth rate of real output, inflation will occur.

1991/AL/II/2

Explain $MV = PY$ as an identity in accounting. It is only when V and Y are constant that a change in M gives rise to an equi-proportional change in P , the money price level. (M/P) is always a constant --- prediction of the classical QTM.

The classical QTM is used in conjunction with the classical theory, which states that (in the long run) flexible relative prices imply full employment. Y in this context refers to full employment output. The classical QTM requires the (long run) assumption of full employment rather than explaining full employment.

1993/AL/II/4

According to the Quantity Theory of Money, $PY = MV$

where P = price level, Y = output, M = money supply and V = velocity.

Assume V is constant. If the percentage change in income is 5% and the percentage change in the money supply is 10%, the inflation rate is (i.e. P increases by) 5% (= 10% - 5%).

(We cannot predict the rate of inflation if V is not constant.)

1999/AL/II/4

(a) Inflation is related to monetary factors in the sense that, other things being equal, prices must rise as money supply is increased because we then have more money to spend on the same quantity of goods. [This can also be explained in terms of the quantity theory, assuming constant velocity and output.]

(b) The quantity equation implies that the inflation rate is equal to the negative of the output growth rate when money supply and velocity are constant. In other words, prices can also rise when we use the same quantity of money to chase after a smaller quantity of goods.

2000/AL/II/8

(a) Using the equation of exchange, we have $MV = PY$. If velocity is constant, nominal income (PY) must grow at the same rate as the money supply. As both V and P can change, we cannot separate the impact on P and Y . If we know how Y grows, we can predict the impact of M on P .

(b) (i) Given velocity is constant, nominal income grows at the same rate as the money supply.

Nominal income growth rate in year 2 = $(110 - 100) / 100 = 10\%$

Nominal income growth rate in year 3 = $(121 - 110) / 100 = 10\%$

(ii) Inflation rate in year 2 = 10%

Inflation rate in year 3 = $10\% - 3\% = 7\%$

(c) If the velocity of circulation is constant and real income grows at a constant known rate, then setting the growth of money supply at a specific rate can achieve the target rate of inflation.

2002/AL/II/2(a)

True.

QTM: $g_P = g_M + g_V - g_Y$

$g_P > 0$ even when $g_M = 0$ as long as $g_V > g_Y$ (such as when velocity is constant and output is declining).

2007/AL/II/3(b)

The velocity of circulation of money is the number of times an average dollar changes hands – as a medium of exchange to facilitate market transactions – within a certain time period.

Using the Equation of Exchange, we can measure the velocity by the ratio of nominal GDP to money supply.

2010/AL/II/2(c)

MV represents the total amount of circulating money in the economy within a given period of time. PY or GDP represents the total money value of (final) goods and services produced in the economy within the same period. $MV = PY$ because, in a monetary economy, money changes hands whenever buyers use the money they hold (supplied by the central bank) to purchase goods and services from the sellers.

The Equation of Exchange implied that:

$$g_P = g_M + g_V - g_Y$$

which implies in turn 2 possible sources of inflation (Quantity Theory of Money as a theory of inflation):

- growth in the money supply --- i.e., $g_M > 0$, given $g_V = g_Y = 0$ (or given $g_M > g_Y$ and $g_V = 0$); and
- output decline --- i.e., $g_Y < 0$, given $g_M = g_V = 0$.

SECTION 6: AGGREGATE DEMAND AND AGGREGATE SUPPLY

(Note: Despite the fact that the AD-AS model was not explicitly required in the CE and AL syllabuses, in tackling a number of CE/AL questions, the AD-AS framework can still be applied.)

6.1 DEMAND- AND SUPPLY-SIDE FACTORS AFFECTING NATIONAL INCOME

MARKING SCHEME

1990/CE/II/37 C	1993/CE/II/38 A	1997/CE/II/44 D	2001/CE/II/30 D	2008/CE/II/28 D (55%)
1991/CE/II/41 B	1993/CE/II/43 D	1997/CE/II/46 B	2004/CE/II/31 A (31%)	2019/DSE/II/27 A
1991/CE/II/45 B	1993/CE/II/51 A	2000/CE/II/34 C	2005/CE/II/30 C (73%)	2019/DSE/II/38 C
1992/CE/II/49 B	1997/CE/II/22 A	2000/CE/II/38 C	2006/CE/II/32 C (48%)	2020/DSE/II/27 C
2020/DSE/II/35 B	2021/DSE/II/36 B	2022/DSE/II/28 D	2022/DSE/II/37 A	2023/DSE/II/28 A
2023/DSE/II/30 A	2025/DSE/II/37 C			

1990/CE/II/3/(b)(i) HK\$ depreciate ⇒ Export prices ↓ ⇒ Export volume ↑ ⇒ Boom in export sector ⇒ Aggregate demand ↑ ⇒ price level ↑	(1) (2) (2) (2) (max: 6)
1990/CE/II/05(b)(ii) ↑ sales tax on all commodities ⇒ prices of all goods ↑ ↓ standard rate ⇒ income after tax of the rich ↑ ⇒ consumption ↑ ⇒ aggregate demand ↑ ⇒ general price level ↑	(2) (4)

1992/CE/II/5(c) (ii) a continuous rise in the general price level	(2)
(iii) Construction of the airport ⇒ ↑ in aggregate demand for resources / demand for workers ⇒ inflation occurs when Hong Kong is near full employment	(2) (2)

1994/CE/II/10(a)(ii)
Local construction workers:
- more job opportunities
- income ↑
Whole economy:
- consumption expenditure ↑ / more business
- GNP ↑
- employment ↑
- social capital ↑
- higher economic growth

2012/DSE/II/13(d)
QTM: $MV = PY$, where M = money stock, V = velocity of circulation of money, P = general price level and Y = real output, assuming V constant. (2)

An increase in M leads to an increase in P when Y increases by a smaller proportion.

OR

An increase in M leads to an increase in P by the same percentage when Y is constant. (2)

2018/DSE/II/7(b)
 $MV = PY$, where M: money supply, V: velocity of circulation of money, P: price level, and Y: output (2)

Assume V and Y are both constant. Then when M increases, P would increase by the same proportion.

OR

Assume V is constant. Then when the percentage increase in M is higher than that in Y, P would increase as well. (2)

2020/DSE/II/4C

C) M = Nominal Money Supply

V = Velocity of Money

P = General price level.

Y = Real National Income

While both V and Y are assumed to be constant in the long run. Percentage change of Money supply equal to the percentage change of price level. Continuously expansionary monetary policy (increase of M) will lead to persistent increase of general price level (increase of P) (4 marks)

2022/DSE/II/6a

- (a) QTM: $MV = PY$, where (4)
M = money supply,
V = velocity of circulation of money, assumed to be constant,
P = price level, and
Y = real output.
% increase in Y being larger than % increase in M would result in a drop in P, i.e. deflation occurs.¹

2023/DSE/DSE/II/12b

- (b) Quantity Theory of Money: $MV = PY$ (1)
M = money supply
V = velocity of circulation of money
P = price level
Y = real output
Assuming V to be constant, (1)
as M increases, P increases if the growth rate of M exceeds that of Y. (1)

1995/CE/I/9(c)

- (i) - consumption expenditure
- investment expenditure
- government expenditure
- exports or net export
- imports or net export

(Mark the **FIRST TWO** points only.)

(1@, max: 2)

- (ii) [Note: the answer must match with the answer to (c)(i)]

Consumption expenditure $\uparrow \Rightarrow$ GDP $\uparrow$

e.g. US residents spent more on food, hotel accommodation, traveling, souvenirs, etc.

OR

Investment expenditure $\uparrow \Rightarrow$ GDP $\uparrow$

e.g. more investment in the provision of transport service, accommodation, food supplies, etc.

OR

Government expenditure $\uparrow \Rightarrow$ GDP $\uparrow$

e.g. US government expenditure on infrastructure related to the event, expansion / maintenance of stadiums, etc.

OR

Net exports $\uparrow \Rightarrow$ GDP $\uparrow$

e.g. more tourists' spending on transport, accommodation, food, clothing, souvenirs, etc.

(Mark the **FIRST TWO** points only.)

(3@, max: 6)

1996/CE/I/12(c)

Demand-side factor:

- government expenditure (in current year) increases

Supply-side factor:

- the labour productivity would be raised

OR

- labour supply would increase because workers originally outside the labour force will be attracted

$\therefore$ GDP (both in current year and in future) would increase.

(Mark the **FIRST** demand-side factor and the **FIRST** supply-side factor only.)

(2)

(2)

(2)

(max: 2)

(1)

1997/CE/I/9(d)(i)

prices of most goods and services after tax $\uparrow$ / cost of production $\uparrow$ / supply of goods and services $\downarrow$

$\therefore$ price level $\uparrow$

(1)

(1)

2004/CE/I/10

After-tax income (i.e. "disposable income") increased.

Consumption expenditure would increase.

$\therefore$ GDP would increase.

(2)

(2)

2005/CE/I/11(a)

Zero tariff encourages more export of goods.

Easier market entry encourages more investment.

Both measures will increase the GDP of Hong Kong.

(2)

(2)

(1)

2007/CE/I/8(b)(ii)

- a larger labour force

- These top-quality Mainland students have a higher productivity.

- These top-quality Mainland students can increase the productivity of the local employees when they work together / brings knowledge and new ideas to Hong Kong.

(1@, max: 2)

[Mark the **FIRST TWO** points only]

2009/CE/I/10(a)

- net exports: more net exports (or more export of services) due to more tourists from mainland China
- consumption: more earnings and income will boost consumption expenditure
- investment: more investment (local and from the Mainland), e.g., more hotels and trading firms.
- government expenditure: more government expenditure, e.g., more on infrastructure to facilitate the trade

(2@, max: 4)

2010/CE/I/9(c)

- (i) - more facilities (attractions) will be constructed $\Rightarrow$ increase in investment expenditure

(2)

- (ii) - more tourists from mainland China and other places $\Rightarrow$ increase in net exports (export of services)

OR

- more local visitors $\Rightarrow$ increase in private consumption expenditure

(2)

2019/DSE/II/10

- (b) Verbal elaboration:

(4)

The middle income group is charged a lower average tax rate, their disposable income rises, consumption and aggregate demand increase, narrowing the deflationary gap.

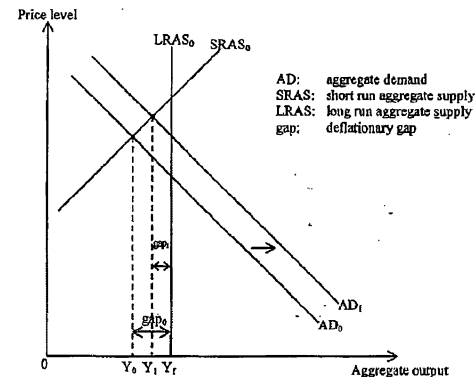
Illustrate in the diagram:

- correct initial deflationary gap
- AD shift rightward
- correct new Y with a smaller gap

(1)

(1)

(1)



For: Pros of the monetary policy OR Cons of the fiscal policy

(2)

- The monetary policy increases Y via investment which will help enhance present/future productivity (while the tax policy may not have such an effect).
- The tax policy may worsen the fiscal balance of the government (while the monetary policy will not have such an effect).
- any other relevant point

Against: Pros of the fiscal policy OR Cons of the monetary policy

(2)

- The tax policy may narrow the income gap (while the monetary policy won't have such an effect).
- Bank loans and money supply may not increase even if there is a fall in the required reserve ratio (while the tax policy will not be affected by the action of commercial banks).
- any other relevant point

D) The investment may include importer raw material which is not the contribution of local resident producing unit.

The investment may include the value of inventories which are not produced in current year.
The investment may include the value of intermediate goods (electricity and other raw materials) which are the contribution of other resident producing unit.

(Mark the first 2 answers) (3 marks)

2023/DSE/II/7b

(b) Verbal elaboration:

- As more travellers visit the country, export of services increases.
- Aggregate demand increases.
- Aggregate output increases, narrowing the deflationary (output) gap.

(1)
(1)
(1)

Indicate in the diagram:

- rightward shift of AD curve
- increase in Y, decrease in deflationary (output) gap

(1)
(1)

6.2 LONG RUN AND SHORT-RUN EQUILIBRIUM

MARKING SCHEME

1994/AL/II/9 A	2012/DSE/II/34 C (51%)	2014/DSE/II/29 D (69%)	2015/DSE/II/39 B (84%)	2018/DSE/II/26 A (80%)
1997/AL/II/21 D	2012/DSE/II/36 D (52%)	2014/DSE/II/34 C (42%)	2016/DSE/II/26 D (65%)	2018/DSE/II/27 D (76%)
1999/AL/II/4 C	2013/DSE/II/23 D (76%)	2014/DSE/II/35 B (67%)	2016/DSE/II/27 C (57%)	2018/DSE/II/28 B (80%)
2000/AL/II/20 D	2013/DSE/II/24 B (65%)	2014/DSE/II/38 D (75%)	2016/DSE/II/28 B (64%)	2018/DSE/II/38 D (58%)
2001/AL/II/14 A	2013/DSE/II/29 A (56%)	2015/DSE/II/26 A (62%)	2017/DSE/II/28 C (76%)	2018/DSE/II/39 A (65%)
2012/AL/II/30 C (79%)	2013/DSE/II/30 D (59%)	2015/DSE/II/27 D (68%)	2017/DSE/II/29 D (60%)	2020/DSE/II/ C
2012/DSE/II/25 A (72%)	2014/DSE/II/27 A (61%)	2015/DSE/II/37 C (59%)	2017/DSE/II/37 B (66%)	2021/DSE/II/27 D
2012/DSE/II/26 A (73%)	2014/DSE/II/28 C (48%)	2015/DSE/II/38 A (73%)	2017/DSE/II/40 B (45%)	2021/DSE/II/28 C
2022/DSE/II/29 B	2022/DSE/II/36 C	2022/DSE/II/38 A	2024/DSE/II/29 B	2024/DSE/II/30 C
2025/DSE/II/28 D	2025/DSE/II/29 A	2025/DSE/II/30 A		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

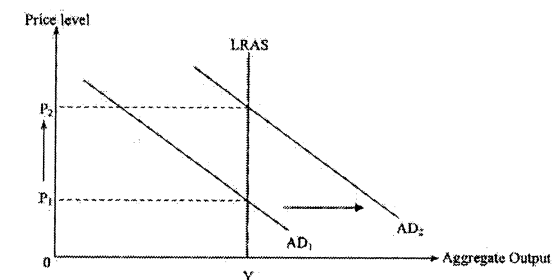
2012/DSE/II/10(c)

People's consumption expenditure increases as their disposable income increases.

Aggregate demand increases.

The price level increases, but aggregate output remains unchanged in the long run.

(2)
(1)
(2)



Indicate in the diagram:

- AD shifts out
- Vertical LRAS
- P increases and Y unchanged

(1)
(1)
(1)